

MA 301
Test 2
Spring - 2013

Problem	points	out of
1		10
2		8
3		16
4		21
5		21
Score		66

Name: _____

HR: _____

Signature: _____

Problem 1: Let C be the boundary of the disc $x^2 + z^2 \leq 4$ in the $y = 1$ plane with positive orientation.

Compute $\oint_C \mathbf{F} \cdot d\mathbf{r}$ for the vector fields: (a) $\mathbf{F} = [5z + 2xz, 1, x^2 + y]$ and (b) $\mathbf{F} = [5zy, x^2, y]$

Problem 2: Let $\mathbf{F} = \left[xz - y, 2yz - z, \frac{1}{2}x^2 + y^2 \right]$ and C be the boundary of the quarter cone: $y^2 = x^2 + z^2$ with $x \geq 0, z \geq 0, 0 \leq y \leq 1$

Compute $\oint_C \mathbf{F} \cdot d\mathbf{r}$

Problem 3: Answer the following conceptual questions:

(a) Suppose the series $\sum_{n=1}^{\infty} a_n$ converges and $a_n > 0$ for all n , determine the convergence of the following:

(i) $\sum_{n=1}^{\infty} \ln(a_n)$

(ii) $\sum_{n=1}^{\infty} \ln(a_n + 1)$

(iii) $\sum_{n=1}^{\infty} \cos(a_n)$

(b) Find the first 3 partial sums of the series $\sum_{n=1}^{\infty} \frac{2^n}{2n}$

(c) (True OR False) If $\sum_{n=1}^{\infty} a_n$ converges, then so does $\sum_{n=100}^{\infty} a_n$

(d) (True OR False) If $\sum_{n=10}^{\infty} a_n$ converges, then so does $\sum_{n=100}^{\infty} a_n$

(e) (True OR False) If $\sum_{n=10}^{\infty} a_n$ converges, then $\lim_{n \rightarrow \infty} a_n = 0$

(b) Suppose $\sum_{k=0}^{\infty} \left(\frac{4c}{5}\right)^k = \frac{5}{c^2}$. Find all values of c which satisfy this expression.

(c) Suppose that the nth partial sum of the series $\sum_{k=1}^{\infty} a_k$ is $s_n = \frac{4n+6}{-1n+10}$.

(i) Find the first 3 partial sums

(ii) Find a_1, a_2, a_3

(iii) Does $\sum_{k=1}^{\infty} a_k$ converge? If so, what is its value?

(e) Write the infinite decimal as a fully reduced fraction:

(b) Express $8.7575757575 \dots$ as a rational number, in the form $\frac{p}{q}$

Problem 4: Determine the convergence of each series. If convergent, find the sum. Justify all work.

(a) $\sum_{n=1}^{\infty} \frac{n^2}{n - n^2}$

(b) $\sum_{n=1}^{\infty} \frac{4}{3^n}$

(c) $\sum_{n=1}^{\infty} \frac{(-5)^{n-1}}{8^n}$

(d) $\sum_{n=3}^{\infty} \frac{\pi^{n+2}}{3^n}$

(e) $\sum_{n=1}^{\infty} \cos\left(\frac{1}{n^2}\right)$

(f) $\sum_{n=0}^{\infty} \frac{3^n}{10^{2n+1}}$

(g) $\sum_{n=4}^{\infty} \ln(n-1) - \ln(n-2)$

Problem 5: Let $z_1 = 1 - i$, $z_2 = 3i$. Compute the following and leave answer in standard form.

(a) $z_1 - iz_2$

(b) $z_1 z_2$

(c) $|z_2|$

(d) $\frac{z_2}{z_1 - z_2}$

Compute the following and leave answer in polar form (preferably Euler's Form: $re^{\theta i}$). No Decimals.

(e) Find the polar form of z_1 .

(f) Find z_1^{12} .

(g) Find $z_1^{1/2}$. (explicitly write out multiple values)

(h) Solve for x : $2x^3 - i = z_2$. (explicitly write out multiple values)